

BAEC ENGINE 1

Capture + Verification

Research and Technical Brief

Project: BAEC Trigger Intelligence

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Software implementation accompanying the conceptual working paper

When the Buyer Is Not In-Market: Buyer-Articulated Evaluation Contingencies in B2B Sales Encounters

Remember what the buyer said mattered, then watch for it.

Abstract

Business-to-business sales encounters often occur before a buyer has decided that evaluating alternatives is worthwhile. The accompanying conceptual working paper defines a buyer-articulated evaluation contingency (BAEC) as information communicated by a presently non-evaluating organizational buyer about a prospective condition that would make initiating or reopening alternative evaluation worthwhile. BAEC Engine 1 is a research software prototype that operationalizes the capture and verification of that information while preserving distinctions among source evidence, model inference, human review, deterministic classification, authorization, and execution. The public demonstration uses a fixed synthetic interaction and one authentic prerecorded model artifact; it makes no model request at runtime and stops after a separate authorization grant is issued. This brief documents the theoretical basis, claim-status hierarchy, software architecture, operational scope, worked synthetic case, provenance, verification status, limitations, and reproducible demonstration protocol. Software verification is evidence about implementation behavior only. It does not validate the BAEC construct, qualify the evaluated model as a default production extraction model, or establish purchase intent.

Source hierarchy and claim status

The conceptual working paper is the theoretical source of truth. The Research Contract translates the paper into software-facing constraints; if the two conflict, the working paper prevails. Implementation behavior and verification artifacts demonstrate software behavior, not empirical validity. The brief uses the claim-status convention below so that theory, managerial interpretation, and software design are not presented as equivalent forms of evidence.

Table 1. Claim-status convention used in this brief.

Status	Meaning	Examples in this brief
DEFINITIONAL	Part of the BAEC or CEE construct definition.	BAEC definition; C1-C4; evaluation activation endpoint.
PROPOSED	A theoretical proposition or future extension that requires empirical testing.	Prospective validity; later monitoring tests.
MANAGERIAL	An account-management interpretation proposed from BAEC information.	Active Opportunity; Conditionally Dormant; No Plausible Path.
IMPLEMENTATION	A software choice made to preserve evidence, authority, or auditability.	AI draft mapping; grant lifetime; separate write-side executor.

1. Research problem and conceptual basis

Many B2B sales interactions occur before a buyer has entered active search or comparison. Two accounts may both reject evaluation today while carrying different information about future relevance. The research question is therefore narrower than generic trigger detection or conditional intention: what salesperson-accessible information becomes observable when a presently non-evaluating buyer identifies a future condition under which alternative evaluation would become worthwhile? (Parks, 2026, §§2.3-3.1).

Buyer-articulated evaluation contingency

An explicit, buyer-generated statement communicated by an organizational buyer who is not currently evaluating relevant alternatives, identifying a prospective condition the buyer expects would make initiating or reopening such evaluation worthwhile.

Parks (2026, §3.1)

BAEC is information communicated in the interaction. It is not a claim that an identical, fully formed rule existed before the conversation, and it is not a commitment to purchase, switch, or select the focal seller. The endpoint is evaluation activation: starting or reopening consideration of relevant alternatives, not purchase or switching (Parks, 2026, §§3.1-3.5).

Conditional evaluation elicitation

Conditional evaluation elicitation (CEE) is the seller behavior through which BAEC-type information may become observable. Following an indication of present non-evaluation, the seller invites the buyer to identify a prospective condition that would make evaluation worthwhile before supplying another seller-generated problem, need, or value argument. CEE is

state-contingent and preserves the buyer, rather than the seller, as the source of the prospective condition (Parks, 2026, §3.6).

2. Constitutive criteria and managerial interpretation

All four BAEC criteria are definitional and constitutive. Each is assessed independently and should be tied to supporting source evidence (Parks, 2026, §§2.4, 3.1).

Table 2. Constitutive BAEC criteria.

ID	Criterion	Requirement
C1	Present non-evaluation	The buyer is presently outside an active evaluation of the relevant alternatives.
C2	Prospective condition	The condition is prospective, not an event that has already occurred.
C3	Buyer articulation	The condition is articulated by the buyer rather than supplied solely by the seller.
C4	Evaluation linkage	The buyer links the condition to initiating or reopening evaluation.

The working paper separately proposes a three-state managerial qualification logic. These labels are account-management interpretations, not validated psychological constructs and not automatic Engine 1 software state transitions (Parks, 2026, §5.1).

Table 3. Managerial three-state interpretation.

State	Meaning	Managerial implication
Active Opportunity	The buyer is evaluating now or is open to evaluation now.	Active selling may be appropriate.
Conditionally Dormant	The buyer is not evaluating now but has articulated a plausible future condition under which evaluation may become worthwhile.	Preserve the condition and reassess selectively rather than repeatedly persuading.
No Plausible Path	No credible future condition is identified, or the condition is too vague, implausible, or not addressable by the seller.	Disengagement may be rational.

A confirmed BAEC does not automatically move an account into Active Opportunity or any other account state. Engine 1 also does not treat a BAEC as current purchase intent, a switching commitment, or standing permission to contact the buyer. Buyer statements can become stale as personnel, contracts, authority, and priorities change.

3. Engine 1 implementation and authority model

Engine 1 operationalizes BAEC capture under an explicit authority model. The engineering objective is not merely to extract language from a conversation. It is to preserve the lineage and authority of each claim as information moves from source interaction, through model-assisted structuring and human verification, into deterministic classification and, only after a separate authorization decision, write-side confirmation.

The architecture therefore maintains several deliberate separations. Source evidence remains primary. Model output remains inference. A human establishes authoritative review decisions. Explicit rules classify the accepted review. Authorization is a distinct human action. Confirmation is performed through a separate write-side execution service. These are IMPLEMENTATION choices designed to preserve the theory-level distinction between what the buyer communicated and what a seller, model, or software layer later inferred.

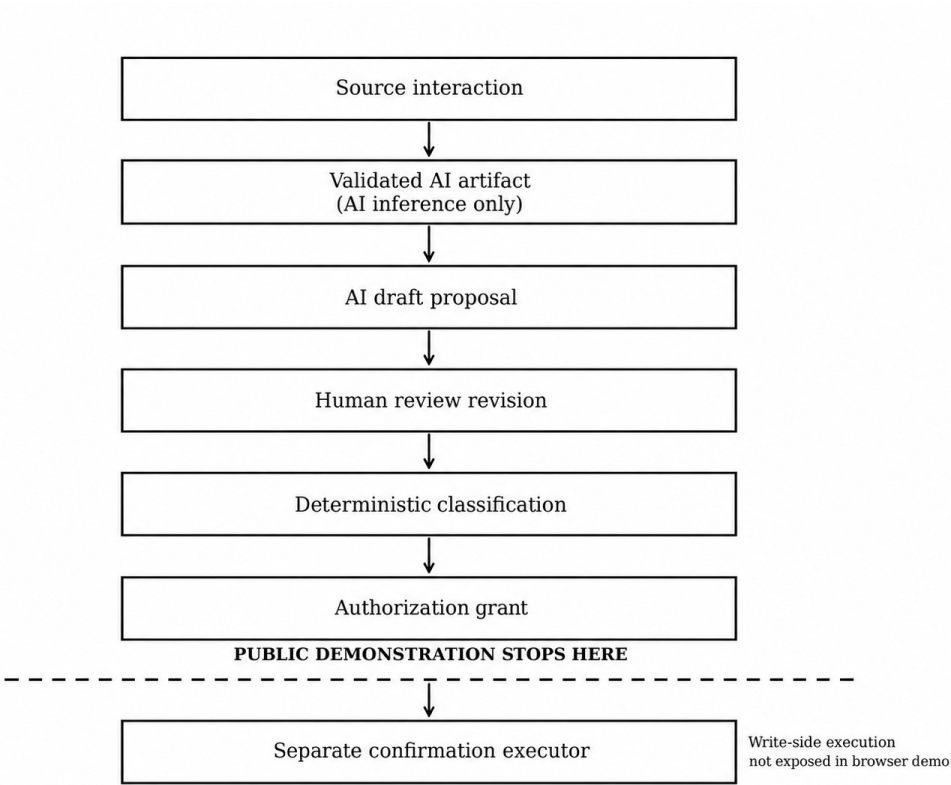


Figure 1. Engine 1 evidence and authority architecture. The public demonstration stops between authorization and the separate confirmation executor.

Operational capability beyond the public demonstration

The underlying Engine 1 implementation is broader than the Streamlit demonstration. Structured account and interaction records can anchor later model runs, artifacts, review revisions, authorization grants, and confirmation lineage. A live extraction path exists with provenance and validation, but the evaluated model was not qualified as the default production extraction model. Human review can select or reject model-proposed evidence, preserve exact threshold semantics, and establish final normalized wording. Once an accepted review is classified, a separate authorization grant can be issued. A write-side Model Context Protocol (MCP) execution service can then revalidate and consume that grant before persisting a confirmed BAEC and its audit linkage. The public browser does not expose that execution capability.

4. Public demonstration and operational scope

The public demonstration is a controlled inspection surface for a subset of Engine 1. It uses one fixed synthetic account interaction, one authentic prerecorded model artifact, isolated per-session in-memory storage, the same human-review and deterministic-classification services used by the implementation, and a separate authorization step. It intentionally stops before confirmation execution.

Table 4. Public demonstration compared with the underlying Engine 1 implementation.

Dimension	Public demonstration	Underlying Engine 1 implementation
Source input	Fixed synthetic Harbor interaction.	Structured accounts and interactions can be persisted and traced.
AI analysis	Authentic prerecorded artifact; no runtime model call.	A live extraction path exists with provenance and validation; the evaluated model is not qualified as the default production model.
Persistence	Temporary per-session in-memory database.	Persistent application storage can maintain audit lineage.
Human review	Interactive evidence and criterion review.	Same authority-bearing review services.

Dimension	Public demonstration	Underlying Engine 1 implementation
Classification	Deterministic classification preview from the accepted review.	Same deterministic BAEC rules.
Authorization	Explicit 15-minute, single-use grant.	Same authorization service and grant constraints.
Execution	Unavailable in the browser demo.	Separate write-side executor can revalidate and consume an authorized grant.

The public interface is deliberately synthetic and non-persistent. It does not accept uploads, customer conversations, API keys, or free-form buyer data. A browser session can record a human review and issue an authorization grant, but it still contains zero confirmed BAECs, zero grant consumptions, and zero confirmation links at the stopping point.

5. Worked synthetic case and model artifact

The demonstration uses a fixed synthetic interaction for Harbor Surgical Center. The example is designed to exercise the full review and authorization path without introducing real customer or patient information.

Synthetic Harbor interaction

Seller: Are you looking at any alternatives for procedure packs right now?

Buyer (Materials Manager): No. We're not looking at other suppliers right now. NorthStar has been fine for us.

Seller: It sounds like there isn't a reason to evaluate alternatives right now. What, if anything, would have to change for that to become worthwhile?

Buyer (Materials Manager): If our supplier raises pricing by more than 10% when our agreement renews, we'd evaluate other options.

The prerecorded artifact identifies two exact excerpts. For readability, Table 5 reconstructs the artifact's evidence-to-criterion mapping. This table is an explanatory rendering of the recorded artifact; it does not assert that the current public interface presents the mapping in this exact table format. All four mappings remain AI inference until a human independently verifies evidence and findings.

Table 5. AI criterion hypotheses mapped to the recorded excerpt IDs.

AI criterion hypothesis	Excerpt ID	Authority status
C1 Present non-evaluation	E1	AI inference only
C2 Prospective condition	E2	AI inference only
C3 Buyer articulation	E2	AI inference only
C4 Evaluation linkage	E2	AI inference only

In the tested successful path, the human reviewer independently verifies E1 and E2 as buyer statements, uses E1 for C1 and E2 for C2-C4, preserves the strict "more than 10%" comparator, and explicitly accepts final normalized wording. The accepted review then satisfies the deterministic BAEC criteria and becomes eligible for separate authorization. The classification does not itself confirm a BAEC or change account state.

Artifact provenance

The Harbor demonstration uses one authentic prerecorded Anthropic response. Retaining the artifact lets every visitor inspect the same structured output without a live model call and lets the loader re-run validation and mapping checks when a demo session is created.

Table 6. Recorded artifact provenance.

Field	Recorded value
Provider	Anthropic
Model	claude-sonnet-5-5
SDK version	anthropic 1.9.0
Prompt version	baec-extraction-prompt/v2
Validator version	baec-extraction-validation/v2
Analysis status	possible_baec_language
Input / output tokens	2914 / 554
Provider message ID	msg_011Cfn2HhMzf73R4csECwf8D
Provider request ID	req_011Cfn2HgE2exXwdCUFGZaV
Canonical recording package digest	8166eccdb4396ba7ccc3f8a01ceebfba56e17fe485abb24b9225e8f402e432a

The package digest above is the SHA-256 of the canonical recording body used by the loader; it is not the SHA-256 of the JSON file bytes. Provenance establishes which source, request, artifact, and versioned validation path were used. It does not establish general model reliability.

6. Verification, limitations, and future boundary

Within the project model-evaluation program (Phase 6), operational integration verification passed, but behavioral eligibility failed. The evaluated model therefore was not qualified as the default production extraction model. The successful Harbor artifact is historical provenance for a synthetic demonstration and does not reverse that decision.

The Version 1.0 software baseline reports 5,818 offline tests passing with one opt-in live extraction evaluation deselected, and public live QA passed. That count is an internal software-verification measure. It is not evidence that the BAEC construct is empirically valid, that the evaluated model is reliable in production, or that a confirmed BAEC predicts revenue, conversion, purchase, switching, or opportunity creation.

The prototype should be interpreted narrowly. The working paper is conceptual and not peer reviewed. The public case is synthetic. Reviewer identity in the demo is self-asserted. Buyer statements can become stale. The browser demo stops after authorization and does not exercise the separate confirmation executor. A future signal may correspond to a verified BAEC condition without establishing that the buyer is currently evaluating or intends to purchase.

Future boundary and Engine 2 roadmap

The working paper proposes buyer-specific monitoring as a later managerial and empirical extension: authorized sources could be monitored for evidence potentially consistent with a condition previously articulated by a particular buyer, followed by human reassessment of the signal, the original contingency, and the appropriateness of re-engagement (Parks, 2026, §§5.2, 6.5-6.6). That monitoring concept is manuscript-grounded. The product label "Engine 2: Monitoring + Correspondence" is a software roadmap term; correspondence functionality is not a claim of the conceptual manuscript and is outside Engine 1.

Engine 1 asks what the buyer said would make evaluation worthwhile and whether that condition can be verified and preserved. Engine 2 is reserved for the later question of whether new evidence may correspond to that buyer-defined condition. Even then, a detected signal should narrow attention rather than automatically create an opportunity or trigger outreach.

Appendix A. Reproducing the successful Harbor demonstration

This appendix records the tested successful path through the public demonstration. The alternate exact-span picker is optional and is not required for this protocol.

Stages 1 and 2

1. Click Start Demo.
2. Read the fixed Harbor interaction and click Continue to AI Draft.
3. Review the prerecorded AI analysis and click Continue to Human Review.

Stage 3A - Verify buyer evidence

4. AI suggestion 1: Use as evidence. It becomes s1.
5. AI suggestion 2: Use as evidence. It becomes s2.
6. Provenance for s1: Verified buyer statement.
7. Provenance for s2: Verified buyer statement.
8. Source selection: s2.
9. Buyer exact statement: Record an exact buyer statement.
10. Selection that is the buyer exact statement: s2.

Stage 3B - Evaluate the four BAEC criteria

Table A1. Human findings for the successful Harbor path.

Field	Entry	Evidence
C1 Present non-evaluation	Criterion met	s1
C2 Prospective condition	Criterion met	s2
C3 Buyer articulation	Criterion met	s2
C4 Evaluation linkage	Criterion met	s2
Articulation origin	Condition stated by buyer	
Elicitation mode	Elicited through an open question	

Stage 3C - Preserve the exact threshold

Table A2. Threshold entries for the successful Harbor path.

Field	Entry
Threshold decision	Record the buyer's threshold
Exact threshold text	more than 10%
Comparator	More than
Number	10
Unit	%
Timing, if needed	Leave blank for the tested Harbor path

The renewal wording remains preserved in the exact source evidence and in the accepted normalized condition.

Stages 3D and 3E - Final wording and submission

11. Use the AI suggestion for the final normalized condition and leave the populated text unchanged.
12. Use the AI suggestion for the final normalized evaluation link and leave the populated text unchanged.
13. Enter a non-identifying reviewer alias such as BP.
14. Confirm that the review checklist reports Ready to submit.
15. Click Accept Review.

Expected Stage 4 result: Human Review - Recorded; Classification Preview - Confirmed BAEC; Authorization - Available.

Then click Continue to Authorization and separately click Authorize BAEC Confirmation. The final screen should report AUTHORIZATION GRANTED, Valid for 15 minutes, Usage - Single use, and Execution - Not yet performed. The browser still contains zero confirmed BAECs, zero grant consumptions, and zero confirmation links.

A useful negative-path check is to mark one criterion Cannot determine or Criterion not met before accepting the review. The expected result is Not a BAEC or Not enough evidence, and authorization remains unavailable. This demonstrates that accepting a human review does not force a BAEC outcome.

Appendix B. Version 1.0 reproducibility snapshot

This appendix records the public Version 1.0 state after live QA. The live-QA software baseline is 2242fe729b4f24a0380a14e7b5879ff1dfd44b1b. The tag baec-engine1-v1.0 will be created after this documentation closeout and is not claimed here as already existing.

Table B1. Version 1.0 reproducibility snapshot.

Item	Current value
Release	Version 1.0 7 October 2026 Public live QA passed
Project	BAEC Trigger Intelligence - Engine 1: Capture + Verification
Public demo	https://baec-engine1.streamlit.app
Public repository	https://github.com/bodhikaiparks/baec-trigger-intelligence
Live-QA software baseline	2242fe729b4f24a0380a14e7b5879ff1dfd44b1b
Primary theoretical source / status	Parks (2026), When the Buyer Is Not In-Market: Buyer-Articulated Evaluation Contingencies in B2B Sales Encounters; unpublished; not peer reviewed
Authentic recording commit	26edcb857aec4a5b0e275187a407ff8ab4ade6f2
Canonical recording package digest	8166eccdb4396ba7ccc3f8a01ceebfba56e17fe485abb24b9225e8f402e432a
Software verification	5,818 passed; 1 opt-in live evaluation deselected
Public demo runtime / data / stop	No live AI; authentic prerecorded artifact; fixed synthetic Harbor interaction; stops at Authorization Granted before confirmation execution
Runtime dependencies	streamlit==1.65.0; pydantic==2.13.5

The software-verification count above is not a research-validation statistic. It measures the automated implementation test suite. The live-QA baseline identifies the application state tested publicly; later documentation-only changes do not alter that software baseline.

References and project artifacts

Parks, B. (2026). When the Buyer Is Not In-Market: Buyer-Articulated Evaluation Contingencies in B2B Sales Encounters. Unpublished conceptual working paper. Not peer reviewed. This working paper contains the full literature review, construct-boundary analysis, propositions, and future research program referenced in this brief.

Working paper availability. The unpublished working paper is not included in this public release. Readers interested in reviewing the manuscript or providing feedback on the BAEC framework may contact the author directly.

BAEC Trigger Intelligence. Research Contract. Internal project artifact translating the working paper into software-facing constraints. The working paper remains the theoretical source of truth.

BAEC Engine 1 authentic Harbor recording and verification package. Synthetic source interaction with one authentic prerecorded model artifact, committed at 26edcb857aec4a5b0e275187a407ff8ab4ade6f2.

Document status: Version 1.0. Public release documented 7 October 2026. Live-QA software baseline: 2242fe729b4f24a0380a14e7b5879ff1dfd44b1b. Planned release tag: baec-engine1-v1.0.